NumPy and Pandas

Numpy Array Operations:

1 — Create the 1D NumPy Array

```
import numpy as np
import pandas as pd

temperatures_w1 = np.array([22.5, 25.3, 20.8, 23.4, 26.1, 24.8, 21.9])

print(temperatures_w1)
```

```
[22.5 25.3 20.8 23.4 26.1 24.8 21.9]
```

2. Inspection and Properties:

```
print("Shape:", temperatures_w1.shape)
print("Data Type:", temperatures_w1.dtype)
print("Number of Elements:", temperatures_w1.size)
```

```
Shape: (7,)
Data Type: float64
Number of Elements: 7
```

3. Array Operations:

```
----> 1 print("Temperatures in Fahrenheit:", temperatures_f)
      2 print("Maximum Temperature:", temperatures_w1.max())
      3 print("Minimum Temperature:", temperatures_w1.min())
      4 print("Mean Temperature:", temperatures_w1.mean())

NameError: name 'temperatures_f' is not defined
```

```
temperatures_f = (temperatures_w1 * 9/5) + 32

print("Temperatures in Fahrenheit:", temperatures_f)
print("Maximum Temperature:", temperatures_w1.max())
print("Minimum Temperature:", temperatures_w1.min())
print("Mean Temperature:", temperatures_w1.mean())
```

```
Temperatures in Fahrenheit: [72.5  77.54 69.44 74.12 78.98 76.64 71.42]
Maximum Temperature: 26.1
Minimum Temperature: 20.8
Mean Temperature: 23.542857142857144
```

4. Array Slicing and Indexing:

```
print("First 3 days:", temperatures_w1[:3])
print("Weekend temperatures:", temperatures_w1[-2:])
print("Middle 3 days:", temperatures_w1[2:5])
```

```
First 3 days: [22.5 25.3 20.8]
Weekend temperatures: [24.8 21.9]
Middle 3 days: [20.8 23.4 26.1]
```

5. Create a 2D Array:

```
temperatures = np.array([
    [22.5, 25.3, 20.8, 23.4, 26.1, 24.8, 21.9],
    [19.2, 22.5, 21.3, 24.0, 23.5, 22.8, 20.1]
])

print(temperatures)
```

```
[[22.5 25.3 20.8 23.4 26.1 24.8 21.9]
 [19.2 22.5 21.3 24.  23.5 22.8 20.1]]
```

6. Inspect and Slice the 2D Array:

```
print("Shape:", temperatures.shape)
print("Data Type:", temperatures.dtype)
print("Total Number of Elements:", temperatures.size)

print("Week 1:", temperatures[0])
print("Week 2:", temperatures[1])

print("Week 1 Weekend:", temperatures[0, -2:])
print("Week 2 Weekend:", temperatures[1, -2:])
```

```
Shape: (2, 7)
Data Type: float64
Total Number of Elements: 14
Week 1: [22.5 25.3 20.8 23.4 26.1 24.8 21.9]
Week 2: [19.2 22.5 21.3 24.  23.5 22.8 20.1]
Week 1 Weekend: [24.8 21.9]
Week 2 Weekend: [22.8 20.1]
```

Pandas Series:

1. **Creating Pandas Series:**

```
marks = pd.Series(
    [95, 92, 89, 85, 80],
    index=['Rank1', 'Rank2', 'Rank3', 'Rank4', 'Rank5']
)

print(marks)
```

```
Rank1    95
Rank2    92
Rank3    89
Rank4    85
Rank5    80
dtype: int64
```

2. **Indexing and Slicing:**

```
print("1st Rank:", marks.iloc[0])

print("\nTop 3 Ranks:")
print(marks.loc[['Rank1', 'Rank2', 'Rank3']])

print("\n3rd Rank:", marks.iloc[2])

print("\nMarks Greater Than 90:")
print(marks[marks > 90])
```

```
1st Rank: 95

Top 3 Ranks:
Rank1    95
Rank2    92
Rank3    89
dtype: int64

3rd Rank: 89

Marks Greater Than 90:
Rank1    95
Rank2    92
dtype: int64
```

3. **Manipulating Series:**

```
marks.loc['Rank1'] = 100

marks = marks.drop('Rank5')

cgpa = marks / 10

print("Updated Marks:")
print(marks)

print("\nCGPA:")
print(cgpa)
```

```
Updated Marks:
Rank1    100
Rank2     92
Rank3     89
Rank4     85
dtype: int64

CGPA:
Rank1    10.0
Rank2     9.2
Rank3     8.9
Rank4     8.5
dtype: float64
```

Pandas DataFrame:

1. **Creating Pandas DataFrame:**

```python
transactions = pd.DataFrame({
    'TransactionID': [101, 102, 103, 104, 105, 106, 107, 108, 109, 110],
    'ProductCategory': [
        'Electronics', 'Clothing', 'Electronics', 'Furniture', 'Clothing',
        'Electronics', 'Furniture', 'Clothing', 'Furniture', 'Electronics'
    ],
    'Region': [
        'North', 'South', 'North', 'East', 'West',
        'North', 'East', 'West', 'South', 'North'
    ],
    'Amount': [200, 150, 300, 450, 200, 250, 300, 180, 350, 400]
})

print(transactions)
```

```
   TransactionID ProductCategory Region  Amount
0            101     Electronics  North     200
1            102        Clothing  South     150
2            103     Electronics  North     300
3            104       Furniture   East     450
4            105        Clothing   West     200
5            106     Electronics  North     250
6            107       Furniture   East     300
7            108        Clothing   West     180
8            109       Furniture  South     350
9            110     Electronics  North     400
```

2. Data Exploration:

```
print("Transactions DataFrame:")
print(transactions)

print("\nFirst 5 Rows:")
print(transactions.head())

print("\nLast 5 Rows:")
print(transactions.tail())

print("\nShape:")
print(transactions.shape)

print("\nColumn Names:")
print(transactions.columns)

print("\nData Types:")
print(transactions.dtypes)
```

```
Transactions DataFrame:
   TransactionID ProductCategory Region  Amount
0            101     Electronics  North     200
1            102        Clothing  South     150
2            103     Electronics  North     300
3            104       Furniture   East     450
4            105        Clothing   West     200
5            106     Electronics  North     250
6            107       Furniture   East     300
7            108        Clothing   West     180
8            109       Furniture  South     350
9            110     Electronics  North     400

First 5 Rows:
   TransactionID ProductCategory Region  Amount
0            101     Electronics  North     200
1            102        Clothing  South     150
2            103     Electronics  North     300
3            104       Furniture   East     450
4            105        Clothing   West     200
```

```
5            106     Electronics  North     250
6            107       Furniture   East     300
7            108        Clothing   West     180
8            109       Furniture  South     350
9            110     Electronics  North     400

First 5 Rows:
   TransactionID ProductCategory Region  Amount
0            101     Electronics  North     200
1            102        Clothing  South     150
2            103     Electronics  North     300
3            104       Furniture   East     450
4            105        Clothing   West     200

Last 5 Rows:
   TransactionID ProductCategory Region  Amount
5            106     Electronics  North     250
6            107       Furniture   East     300
7            108        Clothing   West     180
8            109       Furniture  South     350
9            110     Electronics  North     400

Shape:
(10, 4)

Column Names:
Index(['TransactionID', 'ProductCategory', 'Region', 'Amount'], dtype='str')

Data Types:
TransactionID      int64
ProductCategory      str
Region               str
Amount             int64
dtype: object
```

Display only the 'ProductCategory' and 'Amount' columns.

```
print(transactions[['ProductCategory', 'Amount']])
```

```
  ProductCategory  Amount
0     Electronics     200
1        Clothing     150
2     Electronics     300
3       Furniture     450
4        Clothing     200
5     Electronics     250
6       Furniture     300
7        Clothing     180
8       Furniture     350
9     Electronics     400
```

Retrieve the last 3 columns of the DataFrame using iloc or loc.

```
print(transactions.iloc[:, -3:])
```

```
  ProductCategory Region  Amount
0     Electronics  North     200
1        Clothing  South     150
2     Electronics  North     300
3       Furniture   East     450
4        Clothing   West     200
5     Electronics  North     250
6       Furniture   East     300
7        Clothing   West     180
8       Furniture  South     350
9     Electronics  North     400
```

Filter rows where the 'Region' is 'North' and 'Amount' is greater than 200.

```
north_transactions = transactions[
    (transactions['Region'] == 'North') &
    (transactions['Amount'] > 200)
]

print(north_transactions)
```

```
   TransactionID ProductCategory Region  Amount
2            103     Electronics  North     300
5            106     Electronics  North     250
9            110     Electronics  North     400
```

Find the value counts for the 'ProductCategory' column.

```
print(transactions['ProductCategory'].value_counts())
```

```
ProductCategory
Electronics    4
Clothing       3
Furniture      3
Name: count, dtype: int64
```

Find the unique values in the 'Region' column.

```
print(transactions['Region'].unique())
```

```
<ArrowStringArray>
['North', 'South', 'East', 'West']
Length: 4, dtype: str
```

Group by 'Region' and find the mean amount for each region.

```
region_mean = transactions.groupby('Region')['Amount'].mean()

print(region_mean)
```

```
Region
East     375.0
North    287.5
South    250.0
West     190.0
Name: Amount, dtype: float64
```

3. **Manipulating the DataFrame:**

Modify the 'Amount' for the transaction with TransactionID 102 to 165.

```
transactions.loc[transactions['TransactionID'] == 102, 'Amount'] = 165

print(transactions)
```

```
   TransactionID ProductCategory Region  Amount
0            101     Electronics  North     200
1            102        Clothing  South     165
2            103     Electronics  North     300
3            104       Furniture   East     450
4            105        Clothing   West     200
5            106     Electronics  North     250
6            107       Furniture   East     300
7            108        Clothing   West     180
8            109       Furniture  South     350
9            110     Electronics  North     400
```

Add a new column 'Discount' by calculating 10% of the 'Amount'.

```
transactions['Discount'] = transactions['Amount'] * 0.10

print(transactions)
```

```
   TransactionID ProductCategory Region  Amount  Discount
0            101     Electronics  North     200      20.0
1            102        Clothing  South     165      16.5
2            103     Electronics  North     300      30.0
3            104       Furniture   East     450      45.0
4            105        Clothing   West     200      20.0
5            106     Electronics  North     250      25.0
6            107       Furniture   East     300      30.0
7            108        Clothing   West     180      18.0
8            109       Furniture  South     350      35.0
9            110     Electronics  North     400      40.0
```

- **Remove the row with TransactionID 109.**

```
transactions = transactions[transactions['TransactionID'] != 109]

print(transactions)
```

```
   TransactionID ProductCategory Region  Amount  Discount
0            101     Electronics  North     200      20.0
1            102        Clothing  South     165      16.5
2            103     Electronics  North     300      30.0
3            104       Furniture   East     450      45.0
4            105        Clothing   West     200      20.0
5            106     Electronics  North     250      25.0
6            107       Furniture   East     300      30.0
7            108        Clothing   West     180      18.0
9            110     Electronics  North     400      40.0
```

- **Delete the 'Discount' column from the DataFrame.**

```
transactions = transactions.drop('Discount', axis=1)

print(transactions)
```

```
   TransactionID ProductCategory Region  Amount
0            101     Electronics  North     200
1            102        Clothing  South     165
2            103     Electronics  North     300
3            104       Furniture   East     450
4            105        Clothing   West     200
5            106     Electronics  North     250
6            107       Furniture   East     300
7            108        Clothing   West     180
9            110     Electronics  North     400
```